

Figure 7: For each heuristic, we measure the accuracy of 100 correctly completed prompts associated with a heuristic (blue) and 100 correctly completed prompts not associated with the heuristic (yellow), after ablating that heuristic's neurons. The heuristics are sorted by the accuracy drop induced on associated prompts. Across most heuristics, ablating heuristic neurons causes a larger decrease in accuracy in prompts associated with that heuristic than in not associated prompts.

(200 per layer across 16 layers). Manual inspection of neurons that fail to classify into one of the defined heuristics reveals patterns that are not clearly identifiable (Appendix J).

4.2 Heuristic types are combined to answer arithmetic prompts

Following the classification of the important neurons in each middle- and late-layer MLP into one or several heuristic types, we provide evidence that the bag of heuristics is the primary mechanism the model uses to correctly answer arithmetic prompts. We show this through two ablation experiments.

Knocking out neurons by heuristic type. We now verify that the neurons in each heuristic type contribute to the accuracy of associated prompts by knocking out entire heuristic types and observe the resulting changes in model accuracy. We define a prompt as associated with a heuristic if and only if its components meet the conditions specified by that heuristic. (For instance, the example prompt "$226 - 68 =$" is associated with the heuristic "$op1 \equiv 0 \pmod 2$".) For each heuristic, we sample two sets of 100 correctly completed prompts each, one containing prompts associated with the heuristic and the other containing prompts not associated with that heuristic. For each heuristic, we knock out all neurons classified into it (by setting each $h^{l,n}_{post}$ activation to zero) and then remeasure the accuracy on both sets of prompts. We expect a higher decrease in accuracy on the associated prompts, since we claim each heuristic is causally linked only to its associated prompts.

The results (Figure 7) show that ablating neurons of a specific heuristic causes a significant accuracy drop on associated prompts, more than on not associated prompts, on average. This confirms the causal importance of heuristic neurons in promoting correct answer logits specifically in prompts that are associated with their heuristic type, verifying the targeted functional role of these heuristics.

However, the ablation does not result in a complete accuracy drop; it causes an average drop of 29% out of 95% average pre-ablation accuracy. We find two reasons for this. First, some heuristics have low recall: they do not apply to all associated prompts as they should (see Appendix J). Second, each prompt relies on several unrelated heuristic types, so even when one is ablated, others still contribute to increasing the correct answer's logit. In the following ablation experiment, we verify that this interplay of heuristics provides a fuller image than focusing on one heuristic at a time.

Knocking out neurons by prompt. To provide further evidence that the bag of heuristics is causally linked to correct arithmetic completion, we conduct a second ablation experiment. For each correct prompt, we identify the heuristic types that should affect it based on its operands and ground truth result. We then ablate the neurons with the highest classification scores (Section 4.1) in these heuristics, up to a certain neuron count, and check if the model's completion changes.

The results (Figure 8) show that ablating neurons from associated heuristics significantly drops the model's accuracy, much more than the accuracy drop caused by ablating the same number of randomly chosen neurons from unassociated heuristics. This demonstrates that we can identify the neurons important to a given prompt solely based on its associated heuristics. This also indicates a causal link between the neurons belonging to several heuristics and the prompt's correct completion. This supports our bag of heuristics claim: each heuristic only slightly boosts the correct answer logit, but combined, they cause the model to produce the correct answer with high probability.

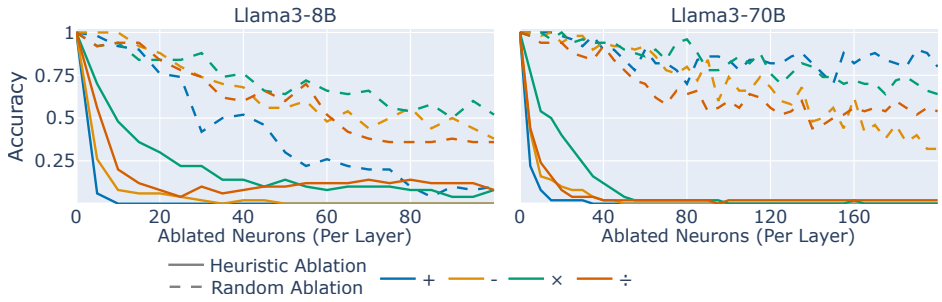


Figure 8: Knocking out neurons that implement heuristics associated with each prompt (full lines) leads to a greater decrease in accuracy than knocking out the same number of neurons whose heuristics are not associated with each prompt (dashed lines). This effect occurs across model sizes.

4.3 Failure modes of the bag of heuristics

The bag of heuristics mechanism employed in Llama3-8B does not generalize perfectly: it fails to achieve perfect accuracy across all arithmetic prompts (Appendix H). This limitation contrasts with the theoretical robustness of a genuine algorithmic approach. Here, we aim to elucidate the specific failures of this mechanism, focusing on *why* it falters for some prompts.

We hypothesize that the bag of heuristics mechanism completes prompts incorrectly in two ways. (1) The "bag" might not be big enough; i.e., a prompt might lack sufficient associated neurons. (2) the heuristics might have imperfect recall (e.g., a neuron that fires for most prompts where the first operand is even, but does not fire for the prompt "$226 - 68 =$") or have low logits for the correct answer token in the value vectors.

To test these hypotheses, we randomly sample 50 correctly completed and 50 incorrectly completed prompts. To test hypothesis (i), we count the number of heuristic neurons associated with each prompt. We find that on average, incorrect prompts have more heuristic neurons associated with them than correct prompts. Therefore, we find no support for this hypothesis. To check hypothesis (ii), we calculate the total contribution of all heuristic neurons to the logit of the correct answer for each prompt. This measurement considers both the specific activation of each neuron for the prompt, as well as the logit of the correct answer token embedded in each neuron's value vector. On average, there is indeed a slight advantage in the total logit contribution for correct prompts over incorrect prompts (Figure 9). This suggests that the primary reason for the bag of heuristics failure on certain prompts is poor promotion of correct answer logits, rather than a lack of heuristics.

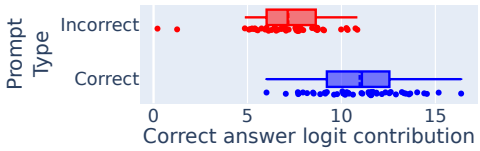


Figure 9: The model's failures can be explained by a lower total logit contribution of the heuristic neurons to the correct answers.

5 Tracking heuristics development across training steps

Does the bag of heuristics emerge as the primary arithmetic mechanism from the onset of training, or does it override an earlier, different mechanism that initially drives arithmetic performance? We conduct an analysis of heuristic development across the training trajectory of the Pythia-6.9B model (Biderman et al., 2023), due to the public availability of its training checkpoints. Specifically, we analyze the model at its final checkpoint (143K steps) and at 10K-step intervals down to 23K steps. The 23K checkpoint is the earliest checkpoint showing good arithmetic performance; Thus, we begin our analysis at this checkpoint. To guide this analysis, we aim to answer three sub-questions:

When do the final heuristic neurons first appear? We examine when each heuristic neuron from the final checkpoint first appears during training. For each neuron classified into a particular heuristic type, we check if the same neuron gets classified into the same heuristic in earlier checkpoints. Averaging this measure across all heuristic types and operators provides insight into when the final heuristics initially appear during training. We observe (Figure 10a) that the model develops its final heuristic mechanism gradually across training, starting from an early checkpoint.

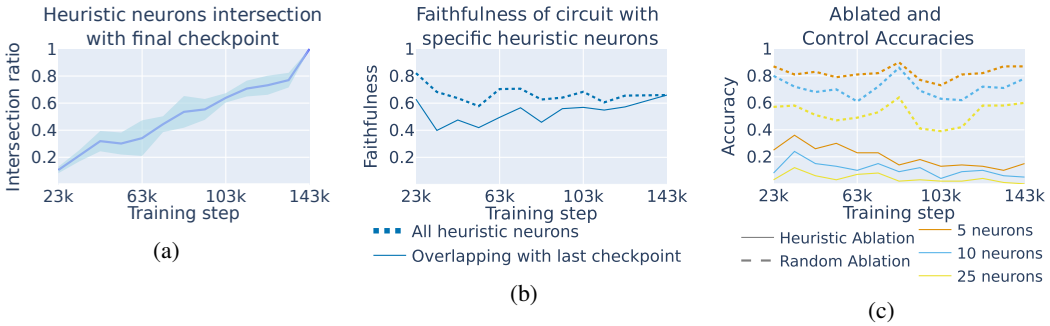


Figure 10: **Heuristic analysis across Pythia-6.9B training checkpoints. (a)** The percentage of heuristic neurons from the last checkpoint that also appear in previous checkpoints increases over training, revealing a gradual creation of the bag of heuristics. **(b)** The heuristic neurons that are mutual with the last checkpoint (full line) explain most of the total heuristic behavior (dashed line) at each checkpoint. Thus, the heuristics that disappear across training are less important to the model. **(c)** Ablating specific heuristic neurons heavily drops the model's accuracy across *all* training checkpoints. This suggests arithmetic accuracy primarily stems from heuristics, even in early stages.

Do additional heuristic neurons exist mid-training? Next, for each mid-training checkpoint, we investigate whether its heuristic neurons that are mutual with the final checkpoint make up the entire heuristic mechanism in that checkpoint, or whether other heuristics exist that later become vestigial. We examine the faithfulness (Section 2.1) of the arithmetic circuit at each checkpoint—once when including only the neurons mutual with the final checkpoint, and once when including all heuristic neurons in that checkpoint. The difference between these two measurements gives us an estimate of the importance of the mutual heuristics. Using this metric, we observe that these final heuristics explain most of the circuit performance for each intermediate checkpoint: they account for an average of 79% of the total heuristics' contribution to accuracy at each checkpoint. This indicates that, while other non-mutual heuristics exist in each checkpoint, these are less important to the circuit's accuracy and slowly become vestigial as the circuit converges to its final form.

Does a competing arithmetic mechanism exist mid-training? Finally, we determine if the heuristics appear as the main arithmetic mechanism from early on in training, or if they co-exist with an unrelated mechanism that becomes vestigial in later checkpoints. We repeat the prompt-guided neuron knockout experiment (Section 4.2) for each checkpoint; i.e., in each checkpoint, we sample 50 correctly completed prompts for each operator. For each prompt, we ablate 5, 10, and 25 heuristic neurons associated with the prompt in that checkpoint. We test if this targeted ablation significantly impairs the model's accuracy, even in earlier stages of training, and compare this to a baseline, where we ablate a similar amount of randomly chosen heuristic neurons. The results (Figure 10c) demonstrate that removing any amount of neurons from heuristics associated with a prompt substantially reduces the model's accuracy on these prompts even at earlier checkpoints, much more than ablating a random set of important neurons. We also observe that ablating 25 heuristic neurons per layer is enough to cause near-zero accuracy in *all* stages of training. This finding asserts that the causal link between a prompt's associated heuristics and its correct completion exists throughout training.

6 Related work

Mechanistic interpretability (MI) aims to reverse-engineer mechanisms implemented by LMs by analyzing model weights and components. Causal mediation techniques (Pearl, 2001) like activation patching (Vig et al., 2020; Geiger et al., 2021), path patching (Wang et al., 2022), and attribution patching (Nanda, 2022; Syed et al., 2023; Hanna et al., 2024b) allow localizing model behaviors to specific model components. Other studies have also presented techniques to explain the effect of specific weight matrices on input tokens (Elhage et al., 2021; Dar et al., 2023), or to analyze activations (nostalgebraist, 2020; Geva et al., 2021). Many studies have aimed to use these techniques to reverse-engineer specific behaviors of pre-trained LMs (Wang et al., 2022; Hanna et al., 2024a; Gould et al., 2024; Hou et al., 2023). We leverage MI techniques to reverse-engineer the arithmetic mechanisms implemented by pre-trained LLMs and explain them at a single-neuron resolution.